

CleanSight

Smart Waste Reporting & Community Cleanup Coordination System

(Personal Portfolio Project – Idea & Design Stage)

1. Introduction

Improper waste disposal and unmanaged garbage dumping are persistent problems in urban and semi-urban areas. Illegal dumping sites, overflowing trash piles, and neglected waste zones contribute to environmental pollution, health risks, and poor living conditions. In many cases, waste-related issues remain unresolved due to delayed reporting, lack of clear evidence, and the absence of centralized tracking systems.

CleanSight is a web-based smart waste reporting and cleanup coordination platform designed to improve how waste issues are identified, verified, and addressed. The system enables citizens to report garbage-related problems by uploading photos and location data. Submitted reports are validated using AI-based image analysis and then displayed on a centralized dashboard.

The platform also allows municipal staff or registered community volunteers to view nearby issues, manage cleanup tasks, and update report statuses. CleanSight focuses on improving visibility, accountability, and response efficiency through technology-assisted community participation.

This project is intended as a **personal portfolio and skill-development project**, demonstrating full-stack development, geolocation handling, AI integration, and real-world problem solving.

2. Project Focus Statement (Core Clarification)

Primary Focus of the Project

The primary focus of CleanSight is the **design and implementation of a smart waste reporting workflow** that combines:

- Image-based issue reporting
- AI-assisted garbage validation
- Location-aware issue tracking
- Task-based cleanup coordination

All other components — including user roles, dashboards, and analytics — are supporting modules that demonstrate how technology can improve waste reporting efficiency and transparency.

The project does **not** aim to function as a nationwide or government-integrated waste management system. Instead, it presents a **realistic, scalable prototype** suitable for small communities, municipalities, or pilot deployments.

3. Problem Statement

Waste management issues often remain unresolved due to several limitations:

- Citizens lack a simple and structured way to report garbage problems
- Reports made via social media or phone calls lack proper tracking
- Authorities receive unclear or false reports without visual evidence
- No centralized platform exists to monitor report status and response progress
- Community volunteers are not effectively connected to nearby cleanup needs

These challenges result in delayed responses, inefficient cleanup coordination, and low accountability. A centralized digital system that integrates image evidence, geolocation, and task tracking can significantly improve waste issue handling.

4. Project Objectives

4.1 Primary Objectives

- Enable citizens to report waste issues using images and location data
- Validate reports using AI-based image analysis
- Provide a centralized dashboard for viewing and managing waste reports
- Support task assignment and status tracking for cleanup activities
- Improve transparency and response visibility for reported issues

4.2 Personal & Technical Objectives

- Demonstrate full-stack web development using modern technologies
 - Implement geolocation and map-based data visualization
 - Integrate AI services for real-world image classification
 - Design scalable database schemas and RESTful APIs
 - Build a production-style prototype suitable for a professional portfolio
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5. System Overview

CleanSight operates as a web-based platform where users can submit waste reports containing images, descriptions, and automatically captured location data. Once submitted, each report undergoes AI-based image validation to confirm the presence of visible garbage.

Validated reports are stored in the database and displayed on a map-based dashboard. Municipal staff or registered volunteers can view nearby issues, accept cleanup tasks, and update the report status upon completion. Citizens can track the progress of their submissions in real time.

The system emphasizes clarity, evidence-based reporting, and structured task coordination rather than automation of physical waste collection.

6. User Roles & Stories

Citizen / Public User

- Upload photos of garbage issues
- Automatically attach location data
- Track report status (pending, in progress, resolved)
- Receive notifications when issues are updated or resolved

Volunteer

- View nearby waste reports
- Accept cleanup tasks
- Upload proof-of-cleanup images
- Track personal contribution history

Municipal Staff

- View all reports within an assigned region
- Assign tasks to volunteers or cleanup teams
- Update and verify report statuses
- Filter reports by urgency, date, or area

Admin

- Manage users and roles
 - Monitor system usage and report statistics
 - Review AI-flagged or invalid reports
 - View overall performance metrics
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7. Key Features

- Image-based waste reporting
 - AI-assisted garbage validation
 - Automatic GPS location tagging
 - Map-based issue visualization
 - Task assignment and status tracking
 - Proof-of-cleanup image uploads
 - Notification system for users and volunteers
 - Admin analytics and monitoring dashboard
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8. Technology Stack

Frontend

- **Next.js** – React-based framework for scalable UI development
- **Tailwind CSS** – Responsive and modern UI styling
- **Mapbox / Google Maps API** – Map-based visualization of reports

Backend

- **Node.js + Express.js** – RESTful API development
- **MongoDB + Mongoose** – Data storage for users, reports, and tasks

Authentication & Storage

- **Firebase Authentication** – Secure user authentication
- **Firebase Storage** – Image storage for reports and cleanup proof
- **Firebase Cloud Messaging (optional)** – Notifications

AI Component

- Pre-trained image recognition API or lightweight ML model
 - Used for validating whether uploaded images contain visible garbage
 - Invalid or unclear reports flagged for manual review
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9. Out of Scope (Explicit Definition)

To ensure feasibility and realistic scope, the following features are excluded:

- Real-time CCTV or live surveillance integration
- Direct integration with government waste systems
- Automated garbage truck dispatch
- Advanced route optimization algorithms
- Native mobile applications

- Full offline-first or PWA implementation
- Financial transactions or payment systems

These features may be considered for future expansion but are not part of the initial implementation.

10. Expected Impact

CleanSight promotes community participation in waste management while improving visibility and accountability for cleanup processes. By combining AI validation, geolocation, and structured task tracking, the platform demonstrates how modern web technologies can support environmental responsibility at a local level.

As a personal portfolio project, CleanSight highlights practical problem-solving, full-stack development skills, and responsible use of AI in a real-world context.

11. Conclusion

CleanSight is a practical, well-scoped project idea suitable for a 5–7 month development timeline. It balances technical complexity with real-world relevance, making it an effective addition to a professional portfolio.

The project demonstrates competency in modern web development, backend API design, AI integration, and system planning without overreaching into unrealistic or enterprise-level promises.
